

2017-Ph-H1-Q13

Section: Our Dynamic Universe

Topic: Gravitation

| Gravitational field strength g decreases with distance from planet.

Use:

$$g = \frac{GM}{r^2}$$

So if r **doubles**, g becomes $\frac{1}{4}$

✅ **Final Answer: C** — $g \propto \frac{1}{r^2}$

Revision Tips:

- Gravity **weakens quickly** with distance
- Always inverse square:

$$g \propto \frac{1}{r^2}, \quad F \propto \frac{1}{r^2}$$